Xi'an Jiaotong-Liverpool University

西交利物浦大学

MODULE CODE	EXAMINER	DEPARTMENT	TEL
CPT101	STEVEN GUAN	COMPUTING	1501

1st SEMESTER 2023/24 RESIT EXAMINATIONS

BACHELOR DEGREE – Year 2

COMPUTER SYSTEMS

TIME ALLOWED: 2 Hours

INSTRUCTIONS TO CANDIDATES

1、 **This is an open book exam.**

2、 **Answer all questions.**

3、 **Answers should be written in the answer booklet(s) provided.**

4、 **No calculator is allowed during the examination.**

5、 **Only answers in English are accepted.**

THIS PAPER MUST NOT BE REMOVED FROM THE EXAM HALL.

Single-Choice Questions

1. Which of the following is system software?
A. Web browser
B. Operating system
C. Word processor

2. The main function of the linker is to:
A. Translate source code into binary
B. Join together compiled modules and resolve external references
C. Execute the program

3. What happens during loading?
A. Source code is translated
B. Executable code is placed into memory for execution
C. Code errors are fixed

4. A compiler translates a program:
A. Line by line while executing
B. As a whole before execution
C. Only when errors occur

5. An interpreter differs from a compiler because it:
A. Translates and executes one instruction at a time
B. Produces an executable file
C. Runs faster than compiled code

6. Which of the following is an example of an interpreted language?
A. C++
B. Java
C. Assembly

7. Why are interpreters sometimes called virtual machines?
A. They emulate hardware by executing code instruction-by-instruction
B. They compile machine code
C. They store virtual memory

8. Dynamic linking allows programs to:
A. Include all code inside one executable
B. Share library code already loaded in memory
C. Be compiled faster

9. Static libraries are:
A. Loaded at runtime
B. Linked during compilation and included in the executable

C. Only for interpreted languages

10. Which of the following is a disadvantage of static libraries?
A. Slow execution
B. Duplicate code copies in memory for each program
C. Difficult linking process

11. The file extension .DLL in Windows refers to:
A. Dynamic Link Library
B. Device Logic Layer
C. Data Link Language

12. When a variable is declared in C, the compiler:
A. Assigns it a memory address and type encoding.
B. Executes it immediately.
C. Converts it into hexadecimal for faster computation.

13. In the onion-ring model of operating systems, the kernel layer:
A. Provides graphical user interaction.
B. Directly controls hardware resources.
C. Stores user-level applications.

14. Which of the following correctly describes multi-tasking?
A. One program finishes before another starts.
B. Multiple programs share CPU time concurrently.
C. One program controls all user input.

15 In a multi-user system, the OS must ensure:
A. Parallel access to the same physical memory space.
B. Fair and protected access to system resources.
C. Unlimited privilege escalation for each user.

Advanced Multiple-Choice Questions
16. Which statements about the binary number system are correct?
A. Each bit represents an increasing power of 2 from right to left.
B. Binary is more compact for humans than decimal.
C. Binary is more suitable for computers due to two-state circuitry.
D. Converting binary to decimal requires bitwise AND operations.
E. Each digit in binary can be 0 or 1.

17. In base-n notation, which statements hold true?
A. Base-10 uses digits 0–9.
B. Base-16 uses digits 0–9 and A–F.

C. Base-8 uses digits 0–7.
D. The base determines the number of unique symbols.
E. Decimal and binary share identical digit sets.

18. For hexadecimal to binary conversion, the following are true:
A. Each hex digit corresponds exactly to 4 binary bits.
B. Conversion requires dividing by 16 repeatedly.
C. The binary group 1111_2 equals F_{16}.
D. Grouping from right to left ensures correct value mapping.
E. Decimal translation is identical to binary translation.

19. Regarding character encoding systems, choose all correct statements:
A. ASCII is a 7-bit standard extended to 8 bits in modern use.
B. EBCDIC is primarily used on IBM mainframes.
C. Unicode can encode characters from multiple writing systems.
D. UTF-8 represents Unicode using variable-length byte sequences.
E. ASCII inherently supports Chinese ideographs.

20. When declaring variables in C or Java, which of the following happen?
A. The compiler allocates memory of the appropriate size.
B. The compiler determines data type representation format.
C. The value is automatically initialized to zero.
D. The symbol table records variable names and addresses.
E. Variables are immediately converted to binary machine code.

21. Which statements about operating system functions are true?
A. Process management allocates CPU resources.
B. Memory management tracks occupied and free blocks.
C. File management provides logical data organization.
D. Device management controls I/O peripherals.
E. User interface management compiles source code.

22. Concerning the command-line interface (CLI) in operating systems:
A. It interprets textual commands from the user.
B. In Unix, it is implemented via shell programs.
C. It directly executes hardware instructions without the OS.
D. It can invoke compilers and utilities.
E. It allows both interactive and batch execution modes.

23. In assembly language, which of the following statements are correct?
A. MOV cannot have both operands as memory locations.
B. MUL implicitly uses EAX as an accumulator.
C. NEG computes the two's complement of its operand.
D. INC and DEC affect status flags.

E. MOVZX performs sign extension of smaller operands.

24. Consider the use of conditional jumps and comparison in x86:
A. CMP sets flags based on a subtraction without storing the result.
B. JE and JZ share the same jump condition.
C. JG checks if the Zero flag is 1.
D. JLE is taken when either less-than or equal condition holds.
E. JNE is taken if Z = 0.

25. Regarding stack operations and function calls, select the correct ones:
A. PUSH stores data at the address pointed to by ESP then decrements ESP.
B. POP copies data from the top of the stack and increments ESP.
C. Parameters for printf are pushed from right to left.
D. CALL automatically pushes the return address.
E. RET pops the return address into EIP and continues execution.

26. Drag-and-drop (for online test) or write the sequence number (for on-site test) of the assembly code to form a program that finds the **sum of even** numbers in an array of **6 integers** and stores this value in the eax register. Note that your sequence must absolutely match the line numbers to the left-most column of the table. The answers for Lines 1,2, and 4 have been provided. Complete the rest.

	Correct Sequence	**Pick From Here**	
Line 1	1	1	`lea esi, myArray`
Line 2	2	2	`mov eax, 0`
Line 3		3	`add eax, [esi]`
Line 4	4	4	`L1: test [esi], 01H`
Line 5		5	`inc eax`
Line 6		6	`L2: inc, esi`
Line 7		7	`add eax, [esi +4]`
Line 8		8	`L2: add esi, 4`
		9	`jnz L2`
		10	`jz L2`
		11	`loop L1`
		12	`loop L2`
		13	`mov ecx, 5`
		14	`mov ecx, 6`

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